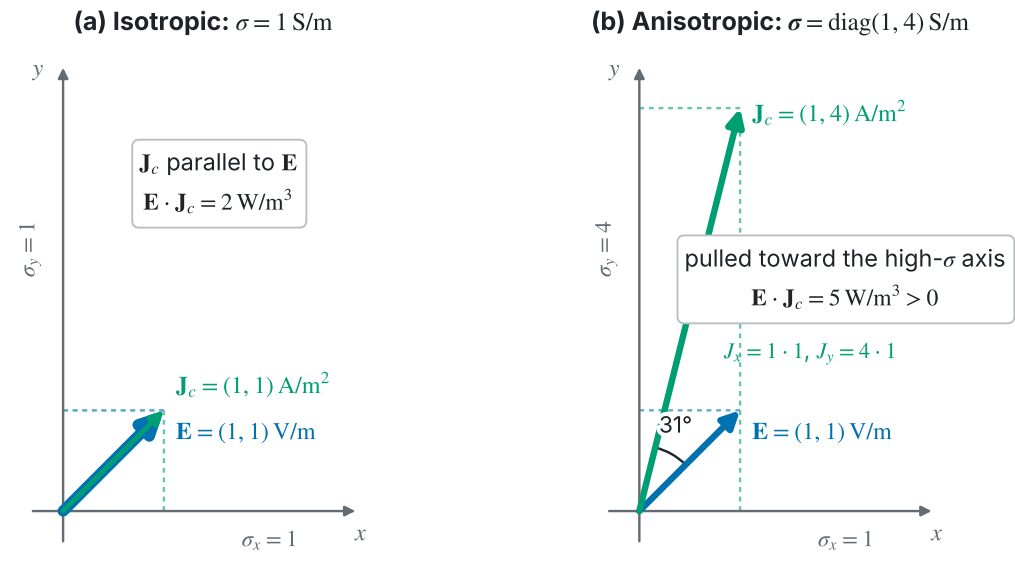
(a) Isotropic: $\sigma = 1$ S/m
y
$\sigma_y = 1$
$\mathbf{J}_c$ parallel to $\mathbf{E}$
$\mathbf{E} \cdot \mathbf{J}_c = 2$ W/m^3
$\mathbf{J}_c = (1, 1)$ A/m^2
$\mathbf{E} = (1, 1)$ V/m
$\sigma_x = 1$
x
(b) Anisotropic: $\sigma = \mathrm{diag}(1, 4)$ S/m
y
$\mathbf{J}_c = (1, 4)$ A/m^2
$\sigma_y = 4$
pulled toward the high-σ axis
$\mathbf{E} \cdot \mathbf{J}_c = 5$ W/m$^3 > 0$
$J_x = 1 \cdot 1$, $J_y = 4 \cdot 1$
31°
$\mathbf{E} = (1, 1)$ V/m
$\sigma_x = 1$
x